

Carlo Francisco “Paco” E. Adajar

Email: paco.adajar@gmail.com

Research Interests

Number theory, especially analytic and elementary. Currently studying questions about the distribution of additive and multiplicative arithmetic functions.

Education

University of Georgia

Ph. D. Mathematics

January 2021–May 2026

Advisor: Paul Pollack

Dissertation: *Three Problems In Analytic And Algebraic Number Theory*

University of the Philippines Diliman

M.S. Mathematics

June 2013–December 2020

Advisor: Richell O. Celeste

July 2016

Thesis: *On odd near-perfect and deficient-perfect numbers with k distinct prime divisors*

Ateneo de Manila University

B.S. Applied Mathematics, Major in Mathematical Finance

March 2013

Work Experience and Positions Held

University of Georgia

Graduate Teaching Assistant

January 2021–May 2026

Ross Mathematics Foundation

Counselor, Ross Mathematics Program

2023, 2024, 2025

Mathematical Society of the Philippines

Test Development Committee, Philippine Mathematical Olympiad

2016–2025

Observer B, Philippine Team to the International Mathematical Olympiad

2018

Deputy Team Leader, Philippine Team to the International Mathematical Olympiad

2020

University of the Philippines Diliman

Lecturer, Institute of Mathematics

August 2020–December 2020

Instructor, Institute of Mathematics

August 2016–July 2020

PAREF Southridge School

Consultant, Mathematics Department

November 2013–March 2015

Mathematics Trainers' Guild of the Philippines

Trainer, Mathematical Olympiad Summer Training Program

April 2012–May 2012

Ateneo de Manila High School

Student Trainer, Department of Mathematics

August 2011–January 2012

Ateneo de Manila Grade School

Student Trainer, Department of Mathematics

August 2010–January 2011

Teaching Experience

University of Georgia

Math 1113: Precalculus

2021–present

Fall 2021

Math 1113: Precalculus ($\times 2$)

Spring 2022

Math 2250: Calculus I

Fall 2022

Math 2250: Calculus I

Spring 2023

Math 2250: Calculus I

Fall 2023

Math 1113: Precalculus ($\times 2$)

Spring 2024

Math 2250: Calculus I

Fall 2024

Math 1113: Precalculus ($\times 2$)

Spring 2025

Math 1113: Precalculus ($\times 2$)

Fall 2025

Math 2250: Calculus I

Spring 2026

David A. Galewski Outstanding Teaching Award

2024

UGA Center for Teaching and Learning Outstanding Teaching Assistant Award

2024

University of the Philippines Diliman

	2016–2020
Math 17: Algebra and Trigonometry ($\times 3$) (discussion class)	1st Semester 2016-2017
Math 53: Elementary Analysis I ($\times 2$) (discussion class)	1st Semester 2016-2017
Math 100: Introduction to Calculus (discussion class)	2nd Semester 2016-2017
Math 53: Elementary Analysis I ($\times 2$) (discussion class)	2nd Semester 2016-2017
Math 54: Elementary Analysis II	Midyear 2017
Math 17: Algebra and Trigonometry (discussion class)	1st Semester 2017-2018
Math 54: Elementary Analysis II ($\times 2$) (discussion class)	1st Semester 2017-2018
Math 53: Elementary Analysis I ($\times 3$) (discussion class)	2nd Semester 2017-2018
Math 21: Elementary Analysis I ($\times 2$) (discussion class)	1st Semester 2018-2019
Math 54: Elementary Analysis II ($\times 2$) (discussion class)	1st Semester 2018-2019
Math 22: Elementary Analysis II ($\times 4$) (discussion class)	2nd Semester 2018-2019
Math 55: Elementary Analysis III (taught for 1/3 of the semester)	2nd Semester 2018-2019
Math 100: Introduction to Calculus	Midyear 2019
Math 22: Elementary Analysis II	Midyear 2019
Math 23: Elementary Analysis III (discussion class)	1st Semester 2019-2020
Math 30: Intermediate Analysis and Applications	1st Semester 2019-2020
Math 54: Elementary Analysis II	1st Semester 2019-2020
Math 21: Elementary Analysis I	1st Semester 2020-2021
Math 23: Elementary Analysis III	1st Semester 2020-2021

Papers

Orders in the Euclidean court. (*with K. J. McGown, P. Pollack, and E. Treviño*)
Submitted.

Books

Institute of Mathematics, University of the Philippines Diliman. *Mathematics 21: Elementary Analysis I Course Module*. (2018) (editor)

E. P. Bautista and I. J. L. Garces. “Mathematical Excursions: A Problem-Solving Primer for Trainers and Olympiad Enthusiasts.” C&E Publishing. (2010) (proofreader)

Selected Talks and Presentations

On the distribution of $v_p(\sigma(n))$ (<i>with P. Pollack</i>) 2025 INTEGERS Conference	May 2025
On the distribution of $v_p(\sigma(n))$ (<i>with P. Pollack</i>) AMS Spring Eastern Sectional Meeting	April 2025
Counting monogenic biquadratic fields Graduate Student Seminar, UGA	February 2025
Unit Fraction Representations Over $\mathbf{Q}$ and $\mathbf{F}_q(t)$ UGA Mathematics Club, UGA	September 2024
Ehrhart theory Graduate Student Seminar, UGA	September 2023
Denser Egyptian fractions Graduate Student Seminar, UGA	September 2022
On statistical properties of prime-divisor counting functions Mock AMS, UGA	August 2022
The worst way to find the area of a circle Graduate Student Seminar, UGA	March 2022
Perfectly imperfect: on odd near-perfect, deficient-perfect, and spoof-perfect numbers Graduate Student Seminar, UGA	February 2021
On odd near-perfect and deficient-perfect numbers ZetaValue2019, Wako, Saitama, Japan	March 2019

Perfect, near-perfect, and deficient-perfect numbers
2016 MSP-NCR First Lecture Series

September 2016

On odd near-perfect and deficient-perfect numbers with k distinct prime divisors (*with R. O. Celeste*)
Poster presented at: 2016 Asian Mathematical Conference

July 2016

On odd near-perfect and deficient-perfect numbers with k distinct prime divisors
2016 MSP Annual Convention

May 2016

Conferences and Research Schools Attended

PANTS XXXVII

University of Georgia, Athens, GA
December 9–10, 2023

PANTS XXXVI

Clemson University, Clemson, SC
October 21–22, 2023

INTEGERS 2023

University of Georgia, Athens, GA
May 17–20, 2023

CTNT 2022

University of Connecticut, Storrs, CT
June 6–12, 2022

100 Years of Mock Theta Functions: New Directions in Partitions, Modular Forms, and Mock Modular Forms
Vanderbilt University, Nashville, TN
May 22–23, 2022

CIMPA School on Group Actions in Arithmetic and Geometry

Universitas Gadjah Mada, Yogyakarta, Indonesia
February 17–28, 2020

Value Distribution of Zeta and L-Functions and Related Topics

Riken, Wako, Saitama, Japan
March 22–26, 2019

VIASM Summer School in Number Theory

Vietnam Institute for Advanced Study in Mathematics, Hanoi, Vietnam
June 18–24, 2018

SEAMS School 2017 Manila: Topics on Elliptic Curves

University of the Philippines Diliman, Quezon City, Philippines
July 16–25, 2017

2016 Asian Mathematical Conference

Bali, Indonesia
July 25–29, 2016

2016 MSP Annual Convention

Cebu, Philippines
May 30–31, 2016

Mentoring

Thesis (co-)adviser (at University of the Philippines Diliman)

Allan James Acong (*with J. T. Dimabayao*) (B.S. Mathematics, 2019)
Mark Joshua Bricenio (*with J. T. Dimabayao*) (B.S. Mathematics, 2019)

Undergraduate mentoring (through the UGA Department of Mathematics Directed Reading Program)

Nicholas Follo (Fall 2021)
Cooper Clickner (Spring 2022)
Maximilian Bennett (Summer and Fall 2022)
Abbie Ottosen (Spring and Fall 2023)
Oliver Sima (Fall 2023)
Alyssa Lovern (Spring 2024)
Nayeong Kweon (Spring 2024)
Stella Sweeney (Fall 2024)
Marley Walter (Fall 2024)

Desmond Yoon (Spring 2025)
Harini Senthil (Fall 2025)
Cash Roberson (Spring 2026)

Service and Extension

Co-organizer and question writer for the UGA High School Math Tournament from 2021–2025.

Co-organizer of the Mathematics Directed Reading Program (DRP) at UGA from Summer 2022–Spring 2026.

Volunteer at the Math Circle at the Athens-Clarke Public Library from Spring 2025–Spring 2026.

Peer mentor for first-year graduate students at UGA from 2022–2024.

Secretary of the AMS Graduate Student Chapter at UGA from 2021–2022.

Co-organized the SEAMS School Manila: Topics on Elliptic Curves held at the University of the Philippines Diliman in 2017.